

TTITLE

Remote Health Monitoring System (RHMS)

COURSE

Software Design and Architecture SWE301

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Introduction

The Remote Health Monitoring System (RHMS) is a software solution that uses IoT technology to monitor patients' vital signs remotely. It allows continuous tracking of health parameters such as heart rate, blood pressure, and body temperature without the need for constant hospital visits. The system helps doctors and healthcare workers access real-time patient data, receive alerts for abnormal readings, and provide timely medical advice. This improves the quality of healthcare delivery, especially for patients with chronic illnesses who require regular monitoring.

The IoT device is a wearable wristband equipped with non-invasive optical glucose sensors and vital sign sensors (heart rate, temperature).

These sensors measure physiological parameters through the skin using infrared and photoplethysmography (PPG) technology.

The readings are transmitted via Bluetooth or Wi-Fi to a central server, where the system checks for abnormal values and alerts the doctor if necessary.

System Overview

Input

- Patient data collected from IoT devices:
 - Heart Rate (HR)
 - Blood Pressure(BP)
 - Temperature (°C)

Output

- Alerts to doctor if readings are abnormal
- Updated patient dashboard with trends and recommendations
- Logs and reports in the system database

System Pipeline / Where it Fits

1. IoT Device Transmit data
2. The data is then transmitted to the server
3. Server analyze the data and Process it
4. Data goes to Doctor's dashboard if abnormal readings are detected
5. Patient Dashboard receives recommendations and follow-up

Design Requirement

Main Component

The main component consists of an IoT device, network for connectivity, cloud server, database, hospital dashboard, doctor's dashboard, patient's dashboard, and alert system. The patient wears the IoT device and it automatically collect vital informations such as blood pressure, heart rate, and temperature. It then send this information to the server. The network which in our case is the internet, enables the IoT device to transmit data, the cloud server receives, stores, and process patient's health data, the database is where patients, doctor, and hospital data will be stored, the hospital dashboard enables the hospital to monitor patient's health data and doctor's data, the doctor's dashboard is a platform where doctor receives patient's report such as abnormal readings and also a platform for the doctor to send feedback, the patient dashboard is where patient recieves doctor's feedback, and the alert system notifies the doctor as soon as abnormal reading is recorded by the IoT device.

User Need

The patient's want to wear a convenient device that automatically record his/her health data and send it to the doctor if abnormality is detected.

Doctor want to receive alert with patient's health data if the readings are abnormal as quick as possible, send feedback as fast as possible and also be able to view health history.

Hospital Administration want to manage doctor and patient information and monitor the overall system performance.

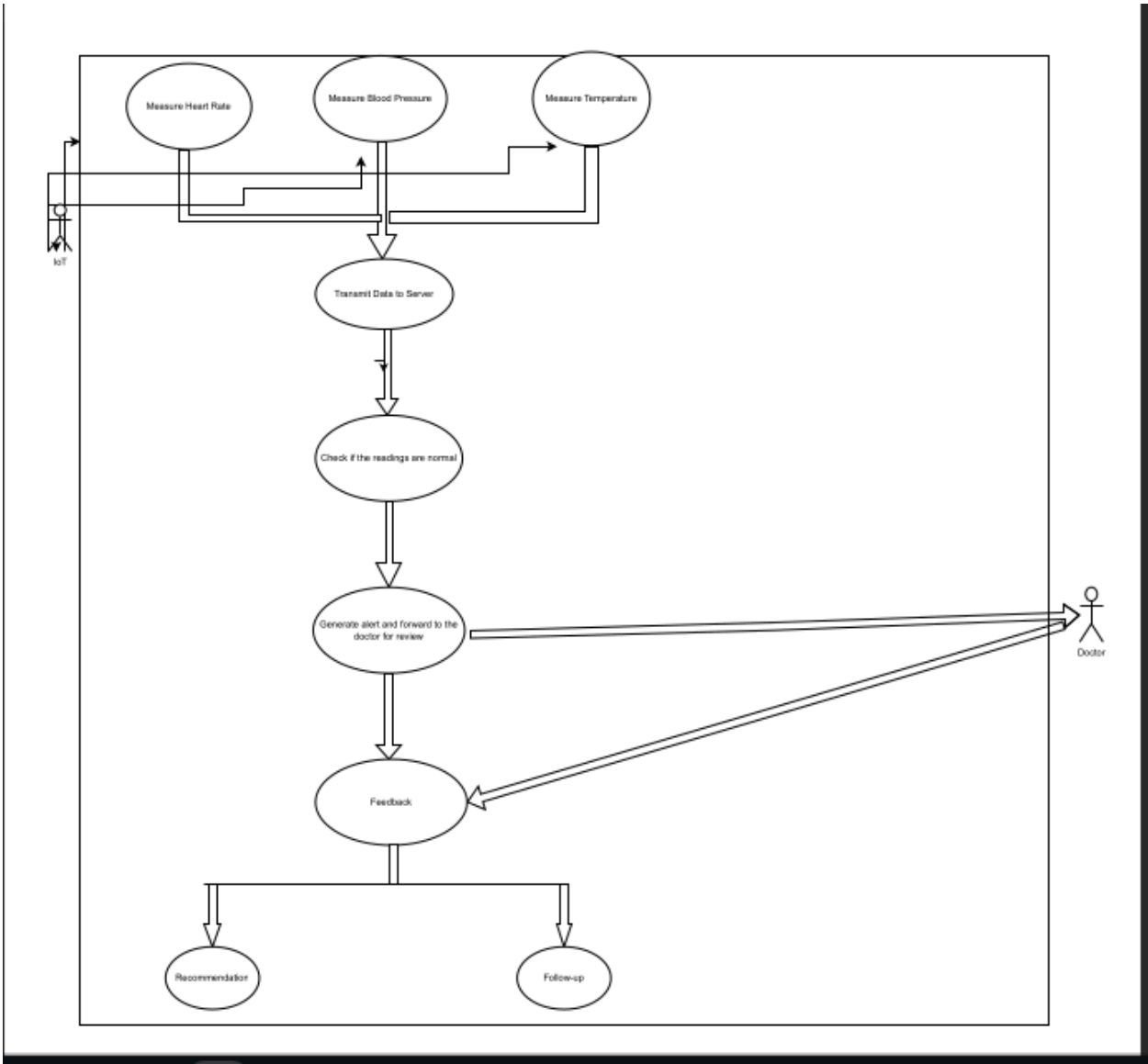
Expected Behavior

The IoT device collect's patient's health data and transmit it periodically, the system analyze the data and check if there is abnormality. If abnormality is detected, the system generate alert and send it to the doctor. The doctor review and analyze the data and provide recommendation, and the system forward this feedback to the patient.

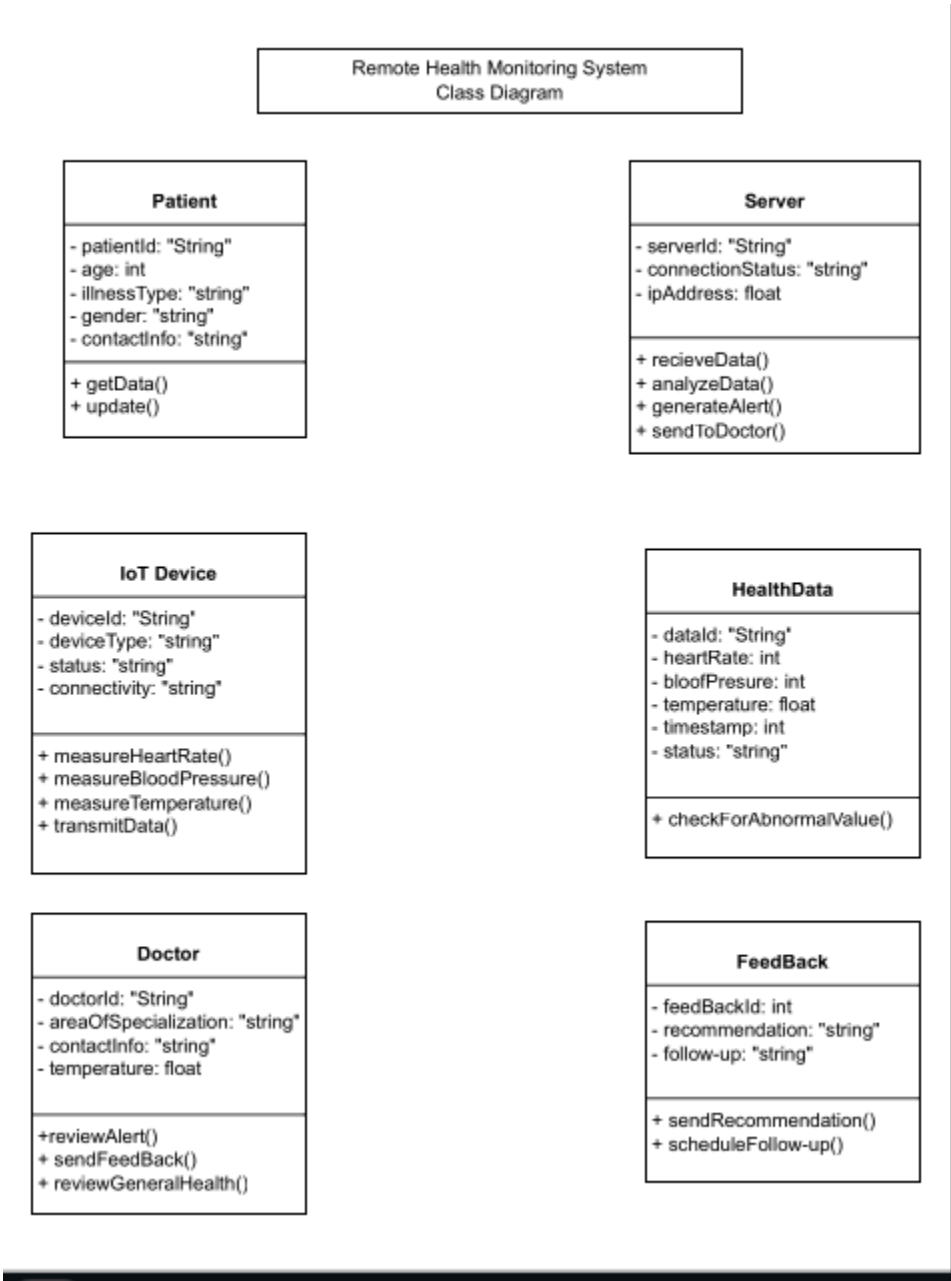
First Design Decision

The system will use client-server architecture with cloud data storage. The IoT device will send data using MQTT(Message Queuing Telemetry Transport) protocol. The programming language to be use is java and python, encryption and authentication will be use for security.

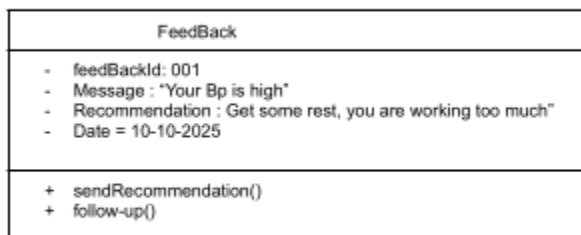
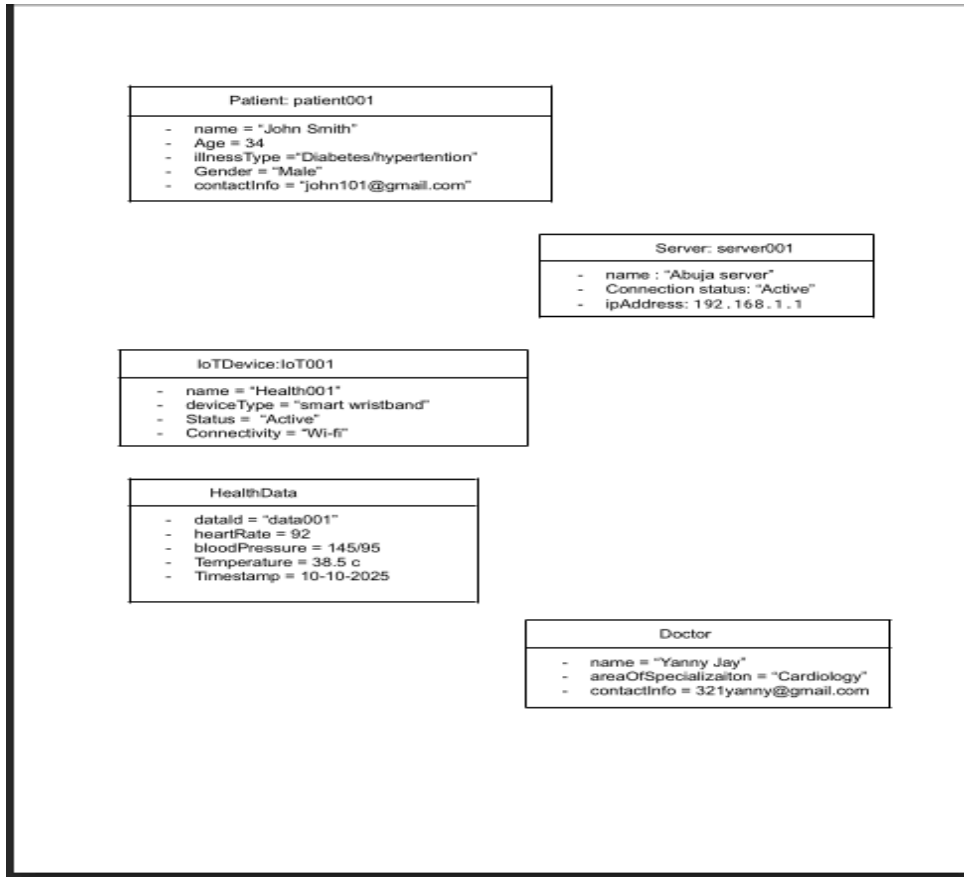
Workflow Description



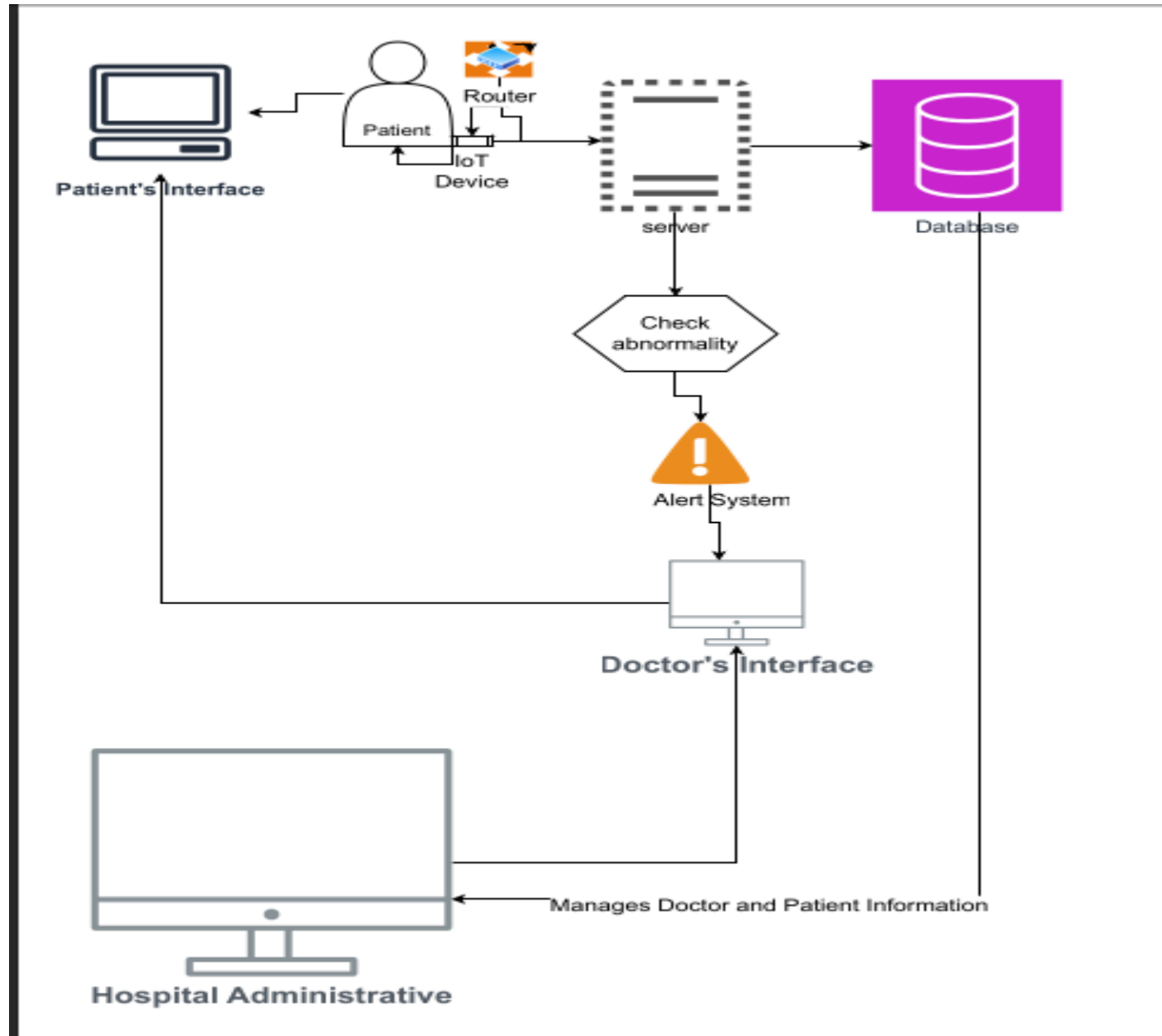
Class Diagram



Object Diagram



Conceptual Diagram



Doctor's Interface

CENTRA LINK HOSPITAL

Alert: Patient's blood pressure reading is 130/85 mmHg — slightly elevated. Monitor closely and follow up if persists



John Smith

As of October 26, 2025

	Recommendation
78 heart rate (HR)	Your blood pressure is slightly elevated at 130/85 mmHg. Continue monitoring your readings daily, maintain a healthy diet low in salt, engage in regular physical activity
130/85 blood pressure (BP)	
37.5 c temperature (Temp)	

SEND

Schedule a meeting



Video meet



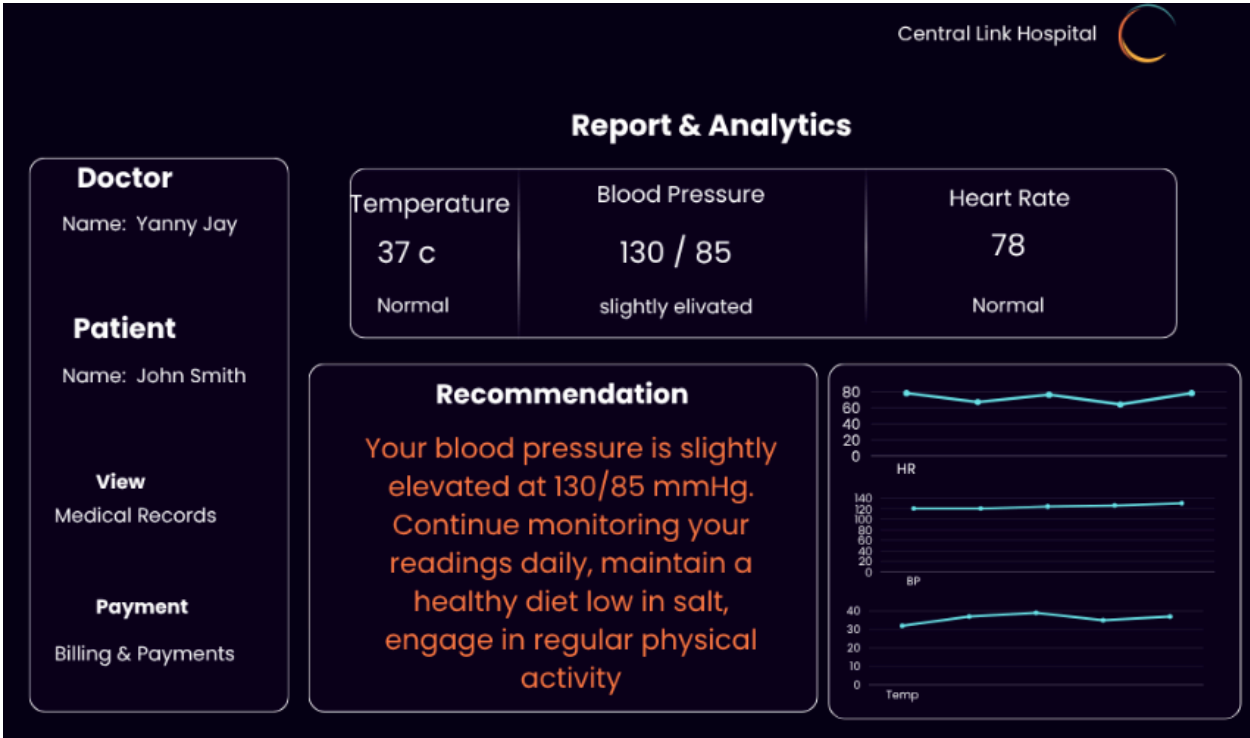
Audio call

Notification

Hospital Management: "All doctors are required to complete the annual training on electronic health records by November 15. Please ensure your credentials are up to date"



Patient's Interface



Document Information

This design document for the Remote Health Monitoring System (RHMS) was prepared by Kyauta Solomon as part of an academic software design project. The current version is 1.0, completed in October 2025. The project is licensed for educational use only and is intended solely for learning and demonstration purposes, not for commercial deployment.

The system relies on several secondary dependencies, including IoT hardware sensors for collecting health data such as heart rate, blood pressure, and body temperature; an active internet connection for real-time data transmission; a web-based dashboard for doctors and patients to access information; and a database system (either cloud-based or local) for securely storing and managing patient records.

Background Information

- Remote health monitoring for chronic diabetic patients
- Importance of early detection of abnormal vitals
- Reduced hospital visits while improving patient outcomes

Actual Procedure / How the System Operates

- Patient wears the IoT device continuously
- Device measures and transmits vitals to the server
- Server processes data, compares with thresholds
- Abnormal readings trigger alerts sent to doctor
- Doctor reviews and sends feedback to patient dashboard

Testing / How to Verify System Works

- **Functionality Testing:** Simulate patient data to see if alerts are triggered correctly
- **Interface Testing:** Verify dashboards update and display correct readings
- **Data Transmission Testing:** Ensure IoT transmit to Server and ensure dashboard workflow is reliable
- **Alert Accuracy Testing:** Check threshold logic for generating alerts

Potential Issues / What Could Go Wrong

- Device battery depletion or malfunction
- Wireless data transmission failure
- Incorrect readings or false alerts
- Server downtime or database errors

Summary

The Remote Health Monitoring System (RHMS) is an IoT-based platform designed to continuously track patients' vital signs such as heart rate, blood pressure, and body temperature. The system transmits this data from wearable IoT devices to a central server, where it is analyzed to detect any abnormal readings. If abnormalities are found, automatic alerts are sent to doctors for review, enabling timely medical intervention. The system also allows doctors to provide feedback, recommendations, and follow-ups remotely, improving healthcare efficiency and patient outcomes.